

## EDUCATION

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| <b>Master of Science in Applied Mathematics</b> , Florida Atlantic University, Boca Raton, FL                                                                                   | 2025–2027 |
| <ul style="list-style-type: none"> <li>GPA: 4.00/4.00</li> <li>Thesis: The Topology of Optimizers at the Edge of Stability</li> <li>Advisor: Dr. Jason Mireles-James</li> </ul> |           |
| <b>Bachelor of Science in Mathematics with Honors</b> , Florida Atlantic University, Boca Raton, FL                                                                             | 2023–2026 |
| <ul style="list-style-type: none"> <li>GPA: 4.00/4.00</li> </ul>                                                                                                                |           |
| <b>High School Diploma</b> , Florida Atlantic University High School, Boca Raton, FL                                                                                            | 2023–2027 |
| <ul style="list-style-type: none"> <li>GPA: 4.00/4.00</li> <li>Concurrently dual-enrolled in combined BS-MS program</li> </ul>                                                  |           |

## AWARDS & HONORS

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| Barry M. Goldwater Scholar (\$7500), Florida Atlantic University              | March 2026   |
| Phi Kappa Phi Honor Society Scholarship (\$1000), Florida Atlantic University | March 2026   |
| Joint Mathematics Meetings Travel Grants (\$250+\$800), Emory University      | January 2026 |
| OURI Research Grant (\$600), Florida Atlantic University                      | January 2026 |
| Florida Launch Scholar (\$610), Florida Atlantic University                   | October 2025 |
| Mu Alpha Theta Summer Grant (\$2000), Florida Atlantic University             | June 2024    |

## RESEARCH EXPERIENCE

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| Massachusetts Institute of Technology, Cambridge, MA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |
| <i>SIG Research Fellow</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jul.–Aug. 2026 |
| <ul style="list-style-type: none"> <li>Selected for competitive six-week research program in geometry processing; completed three two-week projects with researchers and industry professionals</li> <li>Topics: fluid dynamics, computer graphics, computer vision, discrete differential geometry</li> <li><i>Fluid Dynamics on Curved Surfaces (mentor: Dr. Mohammed Sina Nabizadeh):</i> <ul style="list-style-type: none"> <li>Developed Discrete Exterior Calculus-based solvers for incompressible Euler flow on planar and curved triangular meshes, incorporating Hodge projection and Clebsch representations</li> <li>Evaluated stability and mass conservation using perturbed Taylor-Green vortices and Laplace-Beltrami flows</li> <li>Simulated Eulerian flows on Polyscope and Blender</li> <li>Ongoing research; <a href="#">[blog]</a>, <a href="#">[supplementary]</a> in progress.</li> </ul> </li> <li><i>Self-Rectifying Textures (mentor: Dr. Max Dunitz):</i> <ul style="list-style-type: none"> <li>Constructed/deformed textures using homographies and cylindrical warps with OpenCV.</li> <li>Developed Minimum Spanning Tree and Finite Element Method-based rectification pipeline to recover inverse deformation mappings.</li> <li>Work documented in <a href="#">[blog]</a>.</li> </ul> </li> <li><i>Locality in Geometric Algorithms (mentors: Drs. Sebastien Loriot and Mael Rouxel-Labbé):</i> <ul style="list-style-type: none"> <li>Re-implemented Morton, Tipsify, and Hilbert sorting algorithms in C++.</li> <li>Compared runtime of accelerated Edge Collapse algorithm under Morton-sorting, Tipsify-sorting, and Hilbert-sorting against randomly ordered vertices.</li> <li>Work documented in <a href="#">[blog]</a>.</li> </ul> </li> </ul> |                |

Florida Atlantic University, Boca Raton, FL

**Undergraduate Researcher (Advisor: Dr. Yu Xiang)**

2026–Present

- Topics: conformal inference, statistical learning, novelty detection, information theory
- Developed algorithm for conformal novelty detection with finite-sample false discovery rate (FDR) guarantees under a decentralized composite null hypothesis with limited communication.
- Rigorously bounded FDR of my algorithm and empirically demonstrated stronger statistical power and FDR control on synthetic Gaussian numerical experiment in Python.

Emory University, Atlanta, GA

**Computational Mathematics REU (Advisor: Dr. Matthias Chung)**

Jun.–Aug. 2025

- Topics: scientific machine learning, inverse problems, multivariate analysis
- Derived optimal rank-constrained linear encoder-decoder architectures for rank-deficient data and operators.
- Demonstrated competitive performance for inverse recovery tasks on finance and shallow water PDE datasets using PyTorch.
- Contributed theorem development, numerical experiments, and implementation for a collaborative project published in *Foundations of Data Science*.

Florida Atlantic University, Boca Raton, FL

**Master's Thesis (Advisor: Dr. Jason Mireles-James)**

2025–Present

- Topics: deep learning optimization, dynamical systems, computer-assisted proofs
- Derived lower bound for learning rates in which RMSProp and Adam optimization algorithms are Lyapunov unstable and upper bound for attractor size.
- Conducted bifurcation analysis of RMSProp in high learning-rate regimes, characterizing strange attractors and computing the parametrizations and invariant bundles of period-2 invariant tori using MATLAB.
- Validated period-2 orbits, Neimark-Sacker bifurcations, normally hyperbolic quasiperiodic behavior using computer-assisted proofs and interval arithmetic with INTLAB and Julia.

## TEACHING & OUTREACH

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Florida Atlantic University High School, Boca Raton, FL

**Mathematics Tutor**

2023–present

- Tutored 20 students across 400 hours of volunteer tutoring in Algebra II, Precalculus, Calculus, Biology, Physics.
- Mentored 6 students in advanced undergraduate-level mathematics and computer science, providing one-on-one guidance on course selection, internship application essay-writing, connections with faculty in mathematics department.
- Publicized biology lecture notes for 9<sup>th</sup> grade honors biology course [[link](#)].
- Delivered one-on-one and small-group instruction both in person and remotely.

Kyle Loh's Math, YouTube

**Founder and Content Creator**

2022–Present

- Produced undergraduate-level mathematics tutorials with emphasis on differential equations, dynamical systems, and GRE Mathematics Subject topics [[link](#)].
- Developed 137 tutorials reaching approximately 94,000 cumulative views and 730 subscribers.
- Created supplementary instructional tutorials on Google Colab following my ongoing series “Dynamics of Learning Algorithms” designed to improve accessibility of research-level mathematics [[link](#)].

## PUBLICATIONS & PRESENTATIONS

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### *Refereed Articles & Preprints*

- DeLise, **K. Loh**, K. Patel, M. Teague, A. Arnold, M. Chung. "Optimal Linear Baseline Models for Scientific Machine Learning." American Institute of Mathematical Sciences, Foundations of Data Science, 2026. [[doi](#)] [[arXiv](#)]
- **K. Loh**, Y. Xiang. "Decentralized Conformal Novelty Detection via Quantized Model Exchange," 2026. [[arXiv](#)]

### *Poster Presentations*

- Apr 2026. FAU Undergraduate Research Symposium, Florida Atlantic University, Boca Raton, FL. The Convergence Behavior of RMSProp.
- Jan 2026. Joint Mathematics Meetings AMS-PME Student Poster Session, Washington D.C., Optimal Rank-Constrained Mappings for Linear ED Architectures [[pdf](#)].
- Aug 2025. CMDS REU Poster Presentation, Emory University, Atlanta, GA. Optimal Rank-Constrained Mappings for Linear ED Architectures [[pdf](#)].

## PROFESSIONAL MEMBERSHIPS

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| • Phi Kappa Phi Honors Society  | Mar. 2026 |
| • IEEE                          | Mar 2026  |
| • American Mathematical Society | Jan. 2026 |

## TECHNICAL SKILLS

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- **Programming Languages:** Python, MATLAB, Julia, C++, R
- **Machine Learning Frameworks:** PyTorch, TensorFlow, OpenCV
- **Simulation Software:** Blender, Simulink, FEBio
- **Other:** Git, HTML, Astro